

Bharat Agrawal

B.Tech in Computer Science and Engineering

Email: bharat.agrawal1322@gmail.com

[LinkedIn](#) [GitHub](#) [LeetCode](#) [Codeforces](#) [CodeChef](#)

Mobile: +91 9368914189

EDUCATION

Degree/Certificate	Institute/Board	CGPA	Year
Semester Exchange	Technical University of Munich	9.43	2025 - 2026
B.Tech. (CSE)	Indian Institute of Technology Mandi	9.02	2023-Present
Senior Secondary	Kanha Makhan Millennium School, Mathura	94.6%	2023
Secondary	Kanha Makhan Millennium School, Mathura	94.4%	2021

SKILLS

- **Languages:** C/C++, Python, JavaScript, HTML, CSS, SQL
- **Frameworks:** React.js, Next.js, Express.js, Node.js, Flask, FastAPI, Tailwind CSS, Bootstrap
- **Libraries:** Tensorflow, PyTorch, Numpy, Pandas, ScikitLearn, Selenium, BeautifulSoup, Requests
- **Tools:** VSCode, Linux, Git, GitHub, Docker, AWS

PROJECTS

- **Automatic Modulation Recognition** [Github](#)
TensorFlow, Numpy, Ngrok, Sklearn, Flask, MATLAB, VHDL **Dr. Robin Khosla, SCEE, IIT Mandi**
 - **Implementation:** Implemented the **VTCNN2** model using the **Radio ML 2016.10a** Dataset achieved an **overall accuracy of 72%** in classifying **8 Digital** and **3 Analog** signals types of SNR [0,18].
 - **Hybrid Model Development:** Developed a DNN consisting of CNN, LSTM, GRU layers, enhancing the classification accuracy to **92%** for **11 types** of modulated signals.
 - **Improvement:** Reduced model complexity by downsizing to **106K parameters**, decreasing inference latency by **40%** with an average response time of **50 ms**.
- **Tabularis - Table Understanding and Query System** [Github](#)
Python, HTML, Tailwind CSS, JavaScript, Next.js, Flask, Transformers
 - **Implementation:** Developed a pipeline consisting of (Preprocessor - Pix2Struct(240M Params) - Qwen 3.0) to extract tables from PDFs/Images converting them to HTML/CSV Format.
 - **Testing:** Processed 1000+ documents, achieving **95%+ accuracy** in table recognition and data extraction.
 - **Chatbot Integration:** Integrated **Qwen 3.0** LLM with memory, enabling contextual table queries across 50+ diverse query types achieving a 90% query resolution rate.
 - **Achievement:** Secured 2nd place in HCLTech Hackathon (48-hour hackathon) organized in CS671 in our PS.
- **BiometricSEAL - Embedding Biometric in Images** [Github](#)
Python, PyTorch, Vision Transformer, Diffusion, U-Net
 - **Overview:** Modified Meta's VideoSEAL for embedding float watermarks instead of binary watermarks.
 - **Training:** Trained on 60k+ images from **SA-1B** dataset on RTX A5000 GPU. Tried various techniques - freeze-unfreeze, limiting watermarks to 50, Achieved **Cosine Similarities of 0.85 - 0.91** and **SSIM >0.97**.
 - **Model Variant:** Designed a Diffusion-based watermarking model to embed binary watermarks into images achieving **97% Bit Accuracy** on 64-bit watermark.
- **SimuTradex - Paper Trading Website** [Github](#)
Flask, Python, HTML, CSS, JavaScript, Jinja2, Requests, BeautifulSoup, Google Finance
 - **Overview:** Designed a paper trading platform that simulates stock trading with 140+ stock options from NSE with real-time profit and loss tracking, providing users with a realistic trading experience.
 - **Tech Stack and Deployment:** Built using Flask, Python, HTML, CSS, and Jinja2 for seamless user interface providing low-latency execution (<150ms).
 - **Achievement:** Won 1st prize among 100+ freshers in Intracollege GDG KrackHack event.

• Data Harvesting & Structuring Pipeline

Python, Tesseract OCR, Requests, BeautifulSoup, PyPDF2, pdf2image

- **Data Harvesting:** Developed a scalable data harvesting pipeline to autonomously crawl multiple web sources, downloading thousands of documents in PDF, EPUB, and HTML formats.
- **Text Extraction:** Engineered a dual-stage text extraction engine using PyPDF2 and Tesseract OCR, enabling the processing of both text-based and scanned image-based PDFs to ensure maximum data recovery.
- **Efficiency:** Integrated a SHA-256 checksum system to prevent re-processing of unchanged files, boosting pipeline efficiency.

ACHIEVEMENTS

- **1st place** among 100+ freshmen in KrackHack'24 - organized by GDG IIT Mandi.
- Attained **4th** rank in Scythe CTF and **6th** rank in Chronos CTF.
- Qualified for Amazon ML Challenge 2025.
- Ranked **10th nationwide** among 100+ teams in CSAW CTF 2025.
- Solved **1100+** Coding problems across platforms like Codeforces, LeetCode, GFG.

VOLUNTEERING AND PORs

- Core Member, **Kamand Prompt** - Programming Club of IIT Mandi
- Core Member, **Association for Computing Machinery (ACM)** – IIT Mandi Chapter